Multiple testing considerations

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Last time: FWER

Definition: Suppose we test m null hypotheses $H_{0,1}, \ldots, H_{0,m}$. The *family-wise error rate* is the probability of making *at least one* type I error:

$$FWER = P\left(\bigcup_{i:H_{0,i} \text{ is true}} \{\text{reject } H_{0,i}\} \right)$$

<u>Bonferroni</u>: To control FWER at level α, reject $H_{0,i}$ if $p_i < \frac{\alpha}{m}$

Bonferroni is <u>conservative</u>: FWER $\leq \frac{m_0}{m} \alpha$

"Ideal" Bonferroni correction: $\frac{\alpha}{m_0}$

(but don't know m_0!)

Holm's procedure

Suppose we test m null hypotheses $H_{0,1}, \ldots, H_{0,m}$. Let p_i be the corresponding p-value for test i.

- Order the p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
- Let $i^* = \min\left\{i : p_{(i)} > \frac{\alpha}{m-i+1}\right\}$
- Reject $H_{0,(i)}$ for all $i < i^*$

Claim: Holm's procedure controls FWER at level α

<u>Pf</u>: Let $I_0 = \{i : H_{0(i)} \text{ is true}\}$, $m_0 = |I_0|$ (# of true nulls)

Let $j = \min(I_0)$ (index of true null hyp. w/ smallest p-value)

($\forall i < j$, $H_{0(i)}$ is false)

Holm's procedure compares $p_{(j)}$ to $\frac{\alpha}{m-j+1}$

If $p_{(j)} > \frac{\alpha}{m-j+1}$, fail to reject for <u>all</u> true nulls

If $p_{(j)} < \frac{\alpha}{m-j+1}$, make type I error

$P(\text{at least one type I error}) \leq P\left(p_{(j)} < \frac{\alpha}{m-j+1}\right)$

Holm's procedure

Holm's procedure always controls FWER, dominates Bonferroni but often more powerful)

Suppose we test m null hypotheses $H_{0,1}, \ldots, H_{0,m}$. Let p_i be the corresponding p-value for test i.

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cutoff is $\geq \frac{\alpha}{m}$, so can never have less power than Bonferroni

Claim: Holm's procedure controls FWER at level α

$$\text{FWER} = P(\text{at least one type I error}) \leq P\left(p_{(j)} < \frac{\alpha}{m-j+1}\right)$$

There are m_0 elements in I_0 and $j = \min(I_0)$, so

$$m-j+1 \geq m_0$$

$$\Rightarrow \quad P\left(p_{(j)} < \frac{\alpha}{m-j+1}\right) \leq P\left(p_{(j)} < \frac{\alpha}{m_0}\right)$$

$$= P\left(\min_{i \in I_0} p_i < \frac{\alpha}{m_0}\right)$$

$$= P\left(\bigcup_{i \in I_0} \left\{p_i < \frac{\alpha}{m_0}\right\}\right) \underset{\text{(union bound)}}{\leq} m_0 \cdot \frac{\alpha}{m_0} = \alpha$$

$$\Rightarrow \quad \text{FWER} \leq \alpha \qquad //$$

p-value adjustments

May be conceptually nicer to adjust p-value & compare to desired level α (rather than adjusting threshold):

<u>Bonferroni</u>: reject if $p_i < \frac{\alpha}{m} \iff m p_i < \alpha$

Adjusted p-value: $\tilde{p}_i = \min\{m p_i, 1\}$

<u>Holm</u>: $p_{(i)} < \frac{\alpha}{m-i+1} \iff (m-i+1) p_{(i)} < \alpha$

But, only look at $p_{(i)}$ if we reject $H_{0(1)}, \ldots, H_{0(i-1)}$, and want to preserve order

$$\tilde{p}_{(i)} = \min\left\{ \max_{j \le i} \{(m-j+1) p_{(j)}\}, 1 \right\}$$

In R: p.adjust function